

2(a)(i)	(e.g. $a =$) $[16.5 - 0] / [0.30 - 0] = 55 \text{ (m s}^{-2}\text{)}$	A1
2(a)(ii)	$(F =) T - W$	A1
2(a)(iii)	$ma = T - mg$ $m = 16 / (55 + 9.81)$	C1
	$m = 0.25 \text{ kg}$	A1
2(b)(i)	$s = ut + \frac{1}{2}at^2$ or $v^2 = u^2 + 2as$ or $s = vt - \frac{1}{2}at^2$ or $s = \frac{1}{2}(u + v)t$ or $s = \text{area under graph}$ $s = \frac{1}{2} \times 55 \times 0.30^2$ or $s = 16.5^2 / (2 \times 55)$ or $s = 16.5 \times 0.30 - \frac{1}{2} \times 55 \times 0.30^2$ or $s = \frac{1}{2} \times 16.5 \times 0.30$	C1
	$s = 2.5 \text{ m}$	A1
2(b)(ii)	$u = 16.5 \text{ (m s}^{-1}\text{)}$	C1
	$v^2 = u^2 + 2as$ $v^2 = 16.5^2 + 2 \times 9.81 \times 2.5$	C1
	$v = 18 \text{ m s}^{-1}$	A1